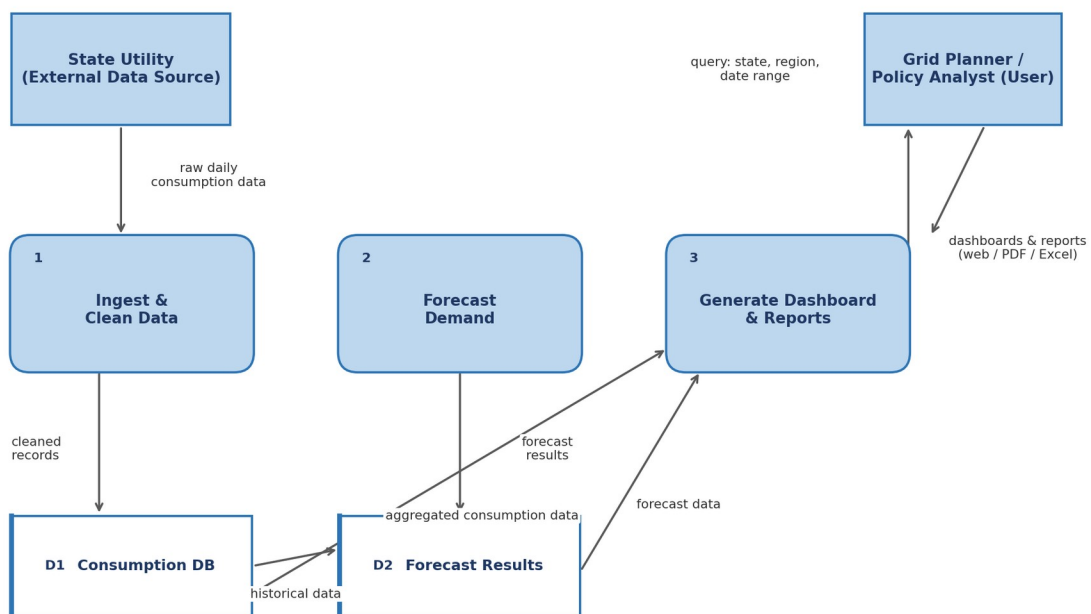


Project Design Phase-II

Data Flow Diagram & User Stories

Date	04 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Data Flow Diagram — Level 0 (Electricity Consumption Analytics Platform)



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Grid Planner (Web user)	Authentication	USN-1	As a grid planner, I can log in with my email and password so that I can access my dashboard.	I can log in and land on my dashboard	High	Sprint-1
		USN-2	As a grid planner, I can reset my password via email if I forget it.	I receive a reset link and can set a new password	Medium	Sprint-1
Grid Planner (Web user)	Data Ingestion	USN-3	As a grid planner, I can upload a daily state-wise consumption CSV file so the platform stays current.	File uploads successfully and appears in the database	High	Sprint-1
System Administrator	Data Ingestion	USN-4	As a system administrator, I can view a validation summary flagging missing or duplicate rows before import is finalized.	Validation report is shown prior to confirming import	High	Sprint-1
Grid Planner (Web user)	Dashboard	USN-5	As a grid planner, I can view a real-time regional consumption dashboard so I can compare NR/WR/SR/ER/NE R regions.	Dashboard loads with all five regions visible and filterable	High	Sprint-2
		USN-6	As a grid planner, I can filter the dashboard by state, region, and date range.	Filtered results update the chart within 3 seconds	Medium	Sprint-2
Grid Planner (Web user)	Forecasting	USN-7	As a grid planner, I can view a 7-day demand forecast for a selected state so I can plan ahead.	Forecast chart displays alongside historical actuals	High	Sprint-2
	Alerts	USN-8	As a grid planner, I receive an automated alert when predicted demand exceeds a set threshold.	Email/in-app alert is sent before the threshold is reached	High	Sprint-3
Policy Analyst (Web user)	Reporting	USN-9	As a policy analyst, I can view year-over-year consumption growth by state to prioritize renewable investment.	Growth-rate chart is available per state, sorted descending	Medium	Sprint-3
		USN-10	As a policy analyst, I can export a forecast report as PDF/Excel to share with my team.	Exported file matches the on-screen data	Low	Sprint-3
System Administrator	Administration	USN-11	As a system administrator, I can	Roles can be assigned and take	Medium	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
			manage user roles and permissions to control access to sensitive data.	effect immediately		